

Commodities

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1 Market Foundation

Core Concept 1.1: Commodity Markets

Commodity markets deal primarily in **physical, tangible assets** that serve as **foundational inputs** for global economic production. Because these raw materials form the base of the manufacturing supply chain, significant price volatility or supply disruptions in primary commodities inevitably cascade down to consumer-end products. *For instance, while wheat represents the underlying commodity traded in financial markets, commercial bread serves as the final manufactured consumer good.*

Modern tradable commodities are systematically classified into three primary sectors.

- **Energy** (Section 4): Primary fuel resources, including crude oil, natural gas, and coal.
- **Metals** (Section 5): Industrial and mineral ores, spanning base, ferrous, and precious varieties.
- **Agriculture** (Section 6): Cultivated assets, comprising grains, oilseeds, soft commodities, and livestock.

Core Concept 1.2: Historical Development of Commodity Trading

Commodity trading represents one of the oldest forms of economic commerce, predating the development of equity and fixed-income markets by several millennia, with historical origins extending back to ancient Mesopotamia.

Early transactions functioned as forward negotiations between individual counterparties to exchange a specific quantity of a commodity at a designated future date—a mechanism known today as an **over-the-counter (OTC)** contract.

However, persistent disagreements regarding quality standards and delivery terms eventually drove the market toward formal standardization, giving birth to centralized futures trading. Early global milestones in this institutional evolution include the **Dojima Rice Exchange in Japan** (1697) and, subsequently, the **Chicago Board of Trade (CBOT)** (1848).

Historically, these centralized marketplaces—along with the subsequent establishment of the **London Metal Exchange (LME) in Europe** (1877)—relied on a physical **open-outcry** system, a traditional framework where floor brokers executed transactions face-to-face within trading pits through synchronized shouting and hand signals.

In the modern era, however, commodity execution has almost entirely transitioned to electronic platforms. As a result of this widespread digital shift, the LME remains unique as one of the final global venues to preserve its legacy physical trading floor.

Core Concept 1.3: Basic Supply & Demand

Commodity markets are fundamentally governed by the core economic principles of **supply and demand** (Section 2). Within this framework, physical storage serves a critical stabilizing function, dampening price volatility and restoring market equilibrium during structural imbalances, such as a sudden supply shock.

Because production centers are rarely geographically aligned with primary consumption hubs, logistics play a defining role in market dynamics. Over 90% of global commodity trade volume is transported via **maritime freight**, with the remainder distributed via pipelines, rail, trucking, and aviation. Consequently, transportation costs represent a significant economic variable that heavily influences trade routes and ultimate market destinations.

Core Concept 1.4: Data Challenges

Comprehensive market analysis requires **vast datasets that are frequently difficult to acquire**. Because sovereign nations report data with varying degrees of frequency, scope, and statistical accuracy, market analysts must dedicate substantial effort to modeling and estimating these baseline values.

2 Commodity Trading Mechanics

2.1 Market Structure & Price

Core Concept 2.1: Physical Pricing and OTC Structure

Commodity valuation requires the **simultaneous evaluation of multiple physical variables**, where variations in quality, grade, and geographic location heavily influence spot pricing.

In the oil sector, for example, **hundreds of distinct crude grades** are traded globally, meaning the limited selection of standardized futures contracts **cannot always accommodate the precise needs of physical market participants**. Consequently, physical commodity commerce remains predominantly structured as a **bilateral business**, directly connecting buyers and sellers via customized OTC transactions. *This contrasts with exchange-based markets, where standardized contracts enable multilateral participation among a large number of buyers and sellers but offer substantially less flexibility in contract design.*

Core Concept 2.2: Reference Pricing and Benchmarking

To establish reliable baseline values amid this operational complexity, the market relies on specialized **Price Reporting Agencies (PRAs)** such as **S&P Global Platts** or **Argus Media**, which publish independent price assessments across a broad spectrum of spot and derivative contracts. These assessments provide critical transparency into the market, allowing the majority of physical OTC transactions to be executed at an agreed price differential relative to these recognized benchmarks.

Furthermore, because **standardized financial futures do not exist for every unique physical grade**, market participants rely on these benchmarks as pricing proxies to manage asset exposure. In practice, a physical OTC agreement for a niche commodity is structured around a pricing formula pegged directly to a dominant benchmark, adjusted by an agreed-upon **quality or location spread**.

Once the physical transaction is linked to the benchmark, the producer or consumer can execute an equal and opposite position in the financial futures market. This strategy creates a dual-market balancing mechanism: any subsequent loss in physical revenue caused by fluctuating spot prices is offset by a corresponding gain in the futures account, and vice versa. Ultimately, while this mechanism requires participants to **forgo the potential upside of favorable price movements**, it **secures operational certainty by locking in baseline profit margins**.

(a) Producer Hedging Against Brent Price Fall

A producer selling a discounted crude grade can contractually peg their physical sales price to Brent minus \$3.00 per barrel while simultaneously shorting Brent financial futures at \$75.00. If the market declines, the reduced revenue from the physical delivery at \$62.00 is precisely counterbalanced by a \$10.00 gain in the short position, thereby securing the targeted net effective return of \$72.00 per barrel.

(b) Refiner Hedging Against Brent Price Rise

Conversely, a refiner purchasing a discounted crude grade can contractually peg their physical acquisition price to Brent minus \$3.00 per barrel while simultaneously executing a long position in Brent financial futures at \$75.00. If the market rallies, the increased procurement cost from the physical delivery at

\$82.00 is precisely counterbalanced by a \$10.00 gain in the long position, thereby securing the targeted net effective purchase cost of \$72.00 per barrel.

Core Concept 2.3: Regional Markets and Arbitrage Dynamics

Crude oil does not trade at a singular, unified global price, operating instead through decentralized regional markets that rely on **localized benchmarks**. Under standard market conditions, the price differentials between these geographic boundaries remain fundamentally constrained; whenever localized imbalances widen these spreads beyond baseline transport costs, market participants exploit the resulting **arbitrage**, shifting physical supply and demand pressures until **price differentials realign**.

Core Concept 2.4: Investment Drivers and Market Financialization

Participation in commodities markets is structurally motivated by a discrete set of investment objectives.

- **Capital preservation** as a safe haven during periods of broader macroeconomic uncertainty
- **Hedging** against inflationary pressures on real asset values
- **Portfolio diversification** through exposure to asset classes exhibiting low correlation with traditional securities
- Direct **speculative positioning** on commodity price movements.

Like traditional financial assets, commodity spot prices are heavily responsive to shifts in broader market sentiment. This behavioral dynamic has become significantly more pronounced alongside the progressive **financialization of commodity markets**, as a growing volume of non-physical participants actively trade contracts.

Core Concept 2.5: Price Drivers

The market valuation of primary commodities is dictated by a diverse array of structural, environmental, and institutional drivers.

(a) Weather & Natural Disasters

Environmental conditions and severe weather anomalies exert direct, immediate shocks on regional commodity supply and demand.

For instance, an unseasonal cold wave in the United Kingdom can rapidly escalate residential heating requirements, triggering a sharp increase in regional natural gas prices due to the sudden demand surge.

(b) Geopolitical Events

Because no sovereign nation is entirely self-sufficient, global economic stability depends heavily on an interconnected network of commodity imports and exports. Consequently, any significant disruption to supply or demand dynamics within key trading nations can precipitate adverse cascading effects throughout the global economy.

This is perfectly exemplified by the surge in crude oil prices stemming from the 2026 Iran-US conflict.

(c) Policy & Regulation

Regulatory mandates and institutional policies frequently reshape demand configurations, directly affecting the profitability of refining and processing firms based on their specific technological and equipment constraints.

A prominent example is the International Maritime Organization (IMO) regulation mandating that all marine vessels utilize fuel with a sulfur content of 0.5% or less, which rapidly altered refinery margins globally by shifting demand toward low-sulfur distillates.

(d) **Demand Elasticity**

Given that primary commodities function as essential societal **necessities**, aggregate consumer demand tends to be **relatively price-inelastic in the short term**, meaning that broader economic slowdowns typically manifest as **gradual, lagging shifts** rather than immediate disruptions.

There are, however, extreme exceptions to this trend. The outbreak of the **COVID-19 pandemic** rapidly decimated nearly one-third of global fuel demand. This unprecedented collapse in near-term consumption forced the crude oil market into an extreme **super contango**, where immediate physical spot prices traded at a dramatic discount relative to longer-dated futures contracts.

2.2 Financialization of Commodities

Core Concept 2.6: Historical Evolution

Core Concept 2.7: Market Sentiment & Speculation

3 Commodity Players

3.1 Producers & Consumers

Core Concept 3.1: Producers & Consumers

3.2 Trading Houses

Core Concept 3.2: Role as Intermediaries

Core Concept 3.3: Arbitrage Strategies

Core Concept 3.4: Major Players

3.3 Investment Banks

Core Concept 3.5: Investment Banks

3.4 Speculators & Investors

Core Concept 3.6: Motivations

Core Concept 3.7: Speculator vs Investor

4 Energy Markets

4.1 Crude Oil Fundamentals

Core Concept 4.1: Brief History of Oil

Core Concept 4.2: Oil Refining and Complexities

4.2 Supply Dynamics

Core Concept 4.3: Seven Sisters & Oil Majors

Core Concept 4.4: Organization of the Petroleum Exporting Countries (OPEC)

Core Concept 4.5: Strategic Petroleum Reserves

4.3 Historical Price Shocks

Core Concept 4.6: Arab Oil Embargo (1973)

Core Concept 4.7: Iranian Revolution (1979)**Core Concept 4.8: Iraq War (1980)****Core Concept 4.9: Price Slump (1980s)****Core Concept 4.10: Price Spike/Fall (1990)****Core Concept 4.11: Asian Economic Crisis (1994)****Core Concept 4.12: Oil Boom (2000s)****Core Concept 4.13: Great Crash (2008)****4.4 Shale Revolution****Core Concept 4.14: Shale Revolution****5 Metal Markets****6 Agricultural Markets**